

MATH 8090: ARMA Models: Inference, Diagnostics, and Model Selection

Readings: CC08 Chapters 7.2–7.5, 8.1, and 6.5; BD16
Chapters 5.2, 5.3, and 5.5; SS17 Chapter 3.5

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Week 6

This Week's Roadmap

- ▶ **Estimate** ARMA parameters using likelihood
- ▶ **Quantify uncertainty** in the parameter estimates
- ▶ **Diagnose** whether the fitted model is adequate
- ▶ **Select** among competing ARMA models

Fit $\longrightarrow$ **Inference** $\longrightarrow$ **Diagnostics** $\longrightarrow$ **Model Selection**

AR(1) Example: Building the Exact Likelihood

Let $\{\eta_1, \eta_2, \dots, \eta_n\}$ be a realization of a zero-mean stationary AR(1) Gaussian time series. Let $\boldsymbol{\theta} = (\phi, \sigma^2)$

$$\ell_n(\boldsymbol{\theta}) = \underbrace{\log(f(\eta_1; \boldsymbol{\theta}))}_{\ell_{n,1}} + \underbrace{\sum_{t=2}^n \log f(\eta_t | \eta_{t-1}, \dots, \eta_1; \boldsymbol{\theta})}_{\ell_{n,2}}.$$

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Note that for $t \geq 2$, $f(\eta_t | \eta_{t-1}, \dots, \eta_1) = f(\eta_t | \eta_{t-1})$, where $[\eta_t | \eta_{t-1}] \sim N(\phi\eta_{t-1}, \sigma^2) \Rightarrow \ell_{n,2} =$

$$-\frac{(n-1)}{2} \log 2\pi - \frac{(n-1)}{2} \log \sigma^2 - \frac{\sum_{t=2}^n (\eta_t - \phi\eta_{t-1})^2}{2\sigma^2}$$

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Also, we know $[\eta_1] \sim N\left(0, \frac{\sigma^2}{(1-\phi^2)}\right) \Rightarrow \ell_{1,n} =$

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AR(1) Log-likelihood Cont'd

$$\ell_n(\boldsymbol{\theta}) = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 + \frac{\log(1 - \phi^2)}{2} - \frac{S(\phi)}{2\sigma^2},$$

where $S(\phi) = \sum_{t=2}^n (\eta_t - \phi\eta_{t-1})^2 + (1 - \phi^2)\eta_1^2$

- For given value of ϕ , $\ell_n(\phi, \sigma^2)$ can be maximized analytically with respect to σ^2

$$\hat{\sigma}^2 = \frac{S(\hat{\phi})}{n}$$

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In the next slides, we are going to contrast **Conditional Least Squares** and **Exact MLE**

Conditional Least Squares vs. Exact MLE

Conditional Least Squares

Condition on η_1 and minimize

$$S(\phi) = \sum_{t=2}^n (\eta_t - \phi \eta_{t-1})^2.$$

$$\text{Set } \frac{dS}{d\phi} = -2 \sum_{t=2}^n \eta_{t-1} (\eta_t - \phi \eta_{t-1}) = 0$$

$$\Rightarrow \hat{\phi}_{\text{CSS}} = \frac{\sum_{t=2}^n \eta_{t-1} \eta_t}{\sum_{t=2}^n \eta_{t-1}^2}.$$

Advantage: simple; closed form.

Key difference: condition on the initial value vs. model its distribution.

Exact MLE

Also model

$$\eta_1 \sim N\left(0, \frac{\sigma^2}{1 - \phi^2}\right).$$

The initial observation therefore contributes to the likelihood.

$\Rightarrow \hat{\phi}_{\text{ML}}$ generally requires numerical optimization.

Advantage: uses the full likelihood.

From AR(1) to General ARMA Models

For the AR(1) model, the likelihood is easy to construct because of the **Markov structure**:

$$f(\eta_t \mid \eta_{t-1}, \dots, \eta_1) = f(\eta_t \mid \eta_{t-1}).$$

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So how can we construct the likelihood efficiently?

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Key Idea: Innovations

Instead of conditioning only on the most recent observation, use the **one-step-ahead prediction errors (innovations)**

Innovations for ARMA Likelihood

Let the **best linear one-step predictor** of η_t be

$$\hat{\eta}_t = \begin{cases} 0, & t = 1, \\ P_{t-1}\eta_t, & t = 2, \dots, n, \end{cases}$$

where $P_{t-1}\eta_t$ denotes the projection of η_t onto the past observations $\eta_1, \dots, \eta_{t-1}$.

- The **one-step prediction errors**, or **innovations**, are

$$U_t = \eta_t - \hat{\eta}_t, \quad t = 1, \dots, n.$$

Their corresponding **innovation variances** are

$$\nu_{t-1} = \mathbb{E}[(\eta_t - \hat{\eta}_t)^2] = \mathbb{E}(U_t^2).$$

- For a causal ARMA process,

$$\nu_{t-1} = \sigma^2 r_{t-1},$$

where r_{t-1} depends only on the AR and MA parameters ϕ and θ , but not on σ^2 .

Likelihood from Innovations

- ▶ **Result 1:** The innovations are independent:

$$U_t \sim N(0, \nu_{t-1}), \quad t = 1, \dots, n.$$

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- ▶ Therefore, the Gaussian log-likelihood is

$$\ell_n(\boldsymbol{\omega}) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^n \log \nu_{t-1} - \frac{1}{2} \sum_{t=1}^n \frac{u_t^2}{\nu_{t-1}}.$$

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- For a causal ARMA process, $\nu_{t-1} = \sigma^2 r_{t-1}$:

$$\ell_n(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2} \sum_{t=1}^n \log r_{t-1} - \frac{1}{2\sigma^2} \sum_{t=1}^n \frac{u_t^2}{r_{t-1}}$$

Maximum Likelihood Estimation for ARMA Models

- ▶ For fixed ϕ and θ , maximizing the likelihood with respect to σ^2 gives

$$\boxed{\hat{\sigma}^2 = \frac{S(\phi, \theta)}{n}}, \quad S(\phi, \theta) = \sum_{t=1}^n \frac{u_t^2}{r_{t-1}}.$$

- ▶ Substituting $\hat{\sigma}^2$ back into the likelihood gives the **profile likelihood** for ϕ and θ :

$$\boxed{(\hat{\phi}, \hat{\theta}) = \arg \max_{\phi, \theta} \ell_{\text{profile}}(\phi, \theta)}.$$

- ▶ In practice, the profile likelihood is maximized using **numerical optimization**.

ARMA Estimation in R: Lake Huron

`arima()` numerically maximizes the likelihood

arima: ARIMA Modelling of Time Series

Description

Fit an ARIMA model to a univariate time series.

Usage

```
arima(x, order = c(0L, 0L, 0L),
      seasonal = list(order = c(0L, 0L, 0L), period = NA),
      xreg = NULL, include.mean = TRUE,
      transform.pars = TRUE,
      fixed = NULL, init = NULL,
      method = c("CSS-ML", "ML", "CSS"), n.cond,
      SSinit = c("Gardner1980", "Rossignol2011"),
      optim.method = "BFGS",
      optim.control = list(), kappa = 1e6)
```

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```

```
```{r}
(MLE_est1 <- arima(lm$residuals, order = c(2, 0, 0), method = "ML"))
```
```

Call:

```
arima(x = lm$residuals, order = c(2, 0, 0), method = "ML")
```

Coefficients:

| | ar1 | ar2 | intercept |
|------|--------|---------|-----------|
| | 1.0047 | -0.2919 | 0.0197 |
| s.e. | 0.0977 | 0.1004 | 0.2350 |

sigma^2 estimated as 0.4571: log likelihood = -101.25, aic = 210.5

Inference for the ARMA Parameters

Motivating example: What is an approximate 95% CI for ϕ_1 in an AR(1) model?

- **Standard errors** can be obtained by computing the inverse of the **Hessian matrix**: $\widehat{\text{Var}}(\hat{\omega}) = -H(\hat{\omega})^{-1}$, where

$$H(\omega) = \frac{\partial^2 \ell_n(\omega)}{\partial \omega \partial \omega^T}$$

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- ▶ For large n , the ML estimators of the ARMA parameters are approximately **jointly normal**:

$$\begin{bmatrix} \hat{\phi} \\ \hat{\theta} \end{bmatrix} \dot{\sim} N \left(\begin{bmatrix} \phi \\ \theta \end{bmatrix}, \frac{V(\phi, \theta)}{n} \right).$$

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- ▶ $V(\phi, \theta)$ is a $(p+q) \times (p+q)$ **asymptotic covariance matrix** (scaled by n) determined by the ARMA parameters.

Inference for ARMA Parameters: Asymptotic Variance

For an AR(1) process,

$$\sqrt{n}(\hat{\phi}_1 - \phi_1) \xrightarrow{d} N(0, 1 - \phi_1^2)$$

or, equivalently, for large n ,

$$\hat{\phi}_1 \dot{\sim} N\left(\phi_1, \frac{1 - \phi_1^2}{n}\right).$$

Thus,

$$\text{SE}(\hat{\phi}_1) \approx \sqrt{\frac{1 - \phi_1^2}{n}}.$$

More generally: For ARMA models, the covariance matrix of the MLEs is typically estimated numerically using the **observed information**:

$$\widehat{\text{Var}}(\hat{\omega}) \approx \left[- \frac{\partial^2 \ell(\omega)}{\partial \omega \partial \omega^T} \Big|_{\hat{\omega}} \right]^{-1}$$

where ω denotes the model parameters.

MLE for Trend and Temporal Correlation in One Step

```
```{r}
(MLE_est4 <- arima(LakeHuron, order = c(2, 0, 0), xreg = yr))
```
```

Call:

```
arima(x = LakeHuron, order = c(2, 0, 0), xreg = yr)
```

Coefficients:

| | ar1 | ar2 | intercept | xreg |
|------|--------|---------|-----------|---------|
| | 1.0048 | -0.2913 | 620.5115 | -0.0216 |
| s.e. | 0.0976 | 0.1004 | 15.5771 | 0.0081 |

sigma^2 estimated as 0.4566: log likelihood = -101.2, aic = 210.4

Fitted model:

$$Y_t = 620.51 - 0.022\text{Year} + \eta_t,$$

where

$$\eta_t = 1.00\eta_{t-1} - 0.29\eta_{t-2} + Z_t, \quad Z_t \sim N(0, \sigma^2 = 0.46).$$

We do not need to estimate the trend first and then fit ARMA to residuals; the regression and dependence parameters can be estimated jointly

Model Diagnostics: What Should We Check?

After fitting an ARMA model, the residuals should contain **no remaining systematic structure**.

1. Remaining dependence?

Residual ACF + Ljung–Box test

2. Changing variance or unusual observations?

Inspect the residual time-series plot.

3. Gaussian assumption reasonable?

Inspect the residual distribution / Q–Q plot.

A good model leaves approximately white-noise residuals.

Diagnostics via the Time Series Residuals

- Recall the innovations are given by

$$U_t = X_t - \hat{X}_t$$

- Under a **Gaussian** model, $\{U_t : t = 1, \dots, T\}$ is an independent set of RVs with

$$U_t \sim N(0, \nu_{t-1}) \stackrel{d}{=} \sigma N(0, r_{t-1}).$$

- Define the **residuals** $\{R_t\}$ by

$$R_t = \frac{U_t}{\sqrt{r_{t-1}}} = \frac{X_t - \hat{X}_t}{\sqrt{r_{t-1}}}$$

Under Gaussian model $R_t \stackrel{i.i.d}{\sim} N(0, \sigma^2)$

ARMA Order Selection

- ▶ Increasing p or q can improve model fit, but also increases *model complexity*.

- ▶ We seek a balance between

good fit and *model parsimony*.

- ▶ *Information criteria* provide a systematic way to compare ARMA models by penalizing unnecessary complexity.

Model Selection: AIC and AICc

The **Akaike Information Criterion (AIC)** balances **model fit** against **model complexity**:

$$\text{AIC} = \underbrace{-2\ell_n(\hat{\theta})}_{\text{lack of fit}} + \underbrace{2k}_{\text{complexity penalty}}$$

where k is the number of estimated model parameters.

Among candidate models fitted to the **same data**,

$$\text{smaller AIC} \implies \text{preferred model.}$$

For smaller samples, the **AICc** adds a stronger finite-sample penalty:

$$\text{AICc} = \text{AIC} + \frac{2k(k+1)}{n-k-1}.$$

Better fit $\longleftrightarrow$ More complexity

AIC/AICc balances the two.

Lake Huron Example: Model Selection

Compare candidate ARMA models using **AIC** and **AICc**:

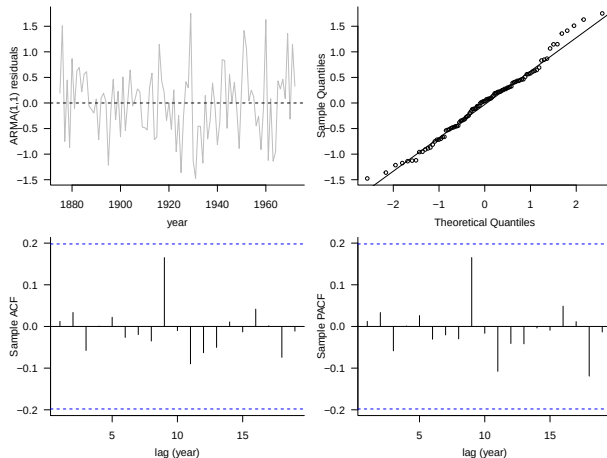
```
m1 <- arima(LakeHuron, order = c(1, 0, 0), xreg = yr)
m2 <- arima(LakeHuron, order = c(1, 0, 1), xreg = yr)
m3 <- arima(LakeHuron, order = c(2, 0, 0), xreg = yr)
m4 <- arima(LakeHuron, order = c(2, 0, 1), xreg = yr)
AIC(m1); AIC(m2); AIC(m3); AIC(m4)
library(MuMIn)
AICc(m1); AICc(m2); AICc(m3); AICc(m4)
`...`
```

```
[1] 218.4501
[1] 212.3954
[1] 212.3965
[1] 214.0638
[1] 218.8803
[1] 213.0476
[1] 213.0487
[1] 214.9868
```

Takeaway: Select the model with the **smallest AIC/AICc**, but then **check its residual diagnostics**.

Model selection $\nrightarrow$ model adequacy

Lake Huron Model Diagnostics



```
> Box.test(resids, lag = 10, type = "Ljung-Box", fitdf = 2)
```

Box-Ljung test

```
data: resids  
X-squared = 3.7882, df = 8, p-value =  
0.8757
```

Putting Everything Together

1. Identify

ACF/PACF + scientific context



2. Estimate

Maximum likelihood



3. Diagnose

Residual plots + residual ACF + Ljung-Box



4. Compare

AIC/AICc + interpretability



5. Use the Model

Inference + forecasting

What About Non-Gaussian Processes?

It is more challenging to express the joint distribution of η_t for non-Gaussian processes. Instead, we often rely on the **Gaussian likelihood** as an **approximate likelihood**

- ▶ In practice:
 - ▶ **Transform** the data to make the series as close to Gaussian as possible (e.g., using a log, square-root, or Box-Cox transformation)
 - ▶ Then use the **Gaussian likelihood** to estimate parameters, assuming the transformed series follows a near-Gaussian structure
 - ▶ For many real-world applications, this approximation works well and simplifies estimation. However, **residual diagnostics** are needed to ensure the model fits the data adequately